

Antonio H. Acuaviva

PHD STUDENT · LANCASTER UNIVERSITY

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Education

Lancaster University

PHD IN MATHEMATICS

Lancaster, UK

October 2023 – Present

- Awarded the **Pure Mathematics Prize** for exceptional performance during the first and third year of the PhD program.
- Researching Functional Analysis, supported by an EPSRC DTP Fellowship.

Cambridge University

MASTER OF ADVANCED STUDY IN MATHEMATICS (PART III)

Cambridge, UK

October 2022 – June 2023

- Awarded with **Distinction**.
- Received the **1912 Senior Scholarship** and **Fitzwilliam College Prize** for academic performance.
- Supported by the prestigious **“la Caixa” Fellowship**.

Universidad Complutense de Madrid (UCM)

BACHELOR'S IN MATHEMATICS AND PHYSICS

Madrid, Spain

September 2017 – September 2022

- **Ranked 1st in Mathematics** (out of 171) and **1st in Physics** (out of 247).
- Recipient of the **Bachelor's Degree Extraordinary Award** in both Mathematics and Physics
- Awarded the “Arturo Duperier” Prize for outstanding performance in Physics.
- Received the **Excellence Grant** every academic year.
- Thesis: *Resilient Source Seeking with Robot Swarms*. Published summary: arXiv:2408.10152.

University of California, Berkeley

UCEAP INTERNATIONAL STUDENT

California, USA

August 2019 – June 2020

- Selected for a year-long study program; recipient of the **UCM-California Fellowship**, awarded to the **top seven students** at Universidad Complutense de Madrid.

Seville International School, San Francisco de Paula

SPANISH AND INTERNATIONAL BACCALAUREATE (IB)

Seville, Spain

September 2015 – June 2017

- Full academic funding provided by a *Goñi and Rey Foundation* Fellowship.

Experience

RESEARCH EXPERIENCE

Fields Institute for Research in Mathematical Sciences

FUSRP: FIELDS UNDERGRADUATE SUMMER RESEARCH PROGRAM

Toronto, Canada

June 2021 – August 2021

- Research internship in mathematical research, collaborating with Professor Ada Chan, Christino Tamon, and Xiaohong Zhang on the development of the theory of Quantum Walks on Graphs.
- Selected to present research results at the CUMC Conference 2021: *Quantum Walks on Graphs: Perfect State Transfer and Periodicity*.
- Results presented in the *Workshop on Algebraic Graph Theory and Quantum Information: Quantum Walks on Hermitian Graphs*.

Institute of Mathematical Sciences (ICMAT)

INTRODUCTION TO RESEARCH GRANT RECIPIENT

Madrid, Spain

June – September 2020 & 2021

- Awarded a competitive “Introduction to Research” grant for **two consecutive years**.
- Collaborated with senior researchers on mathematical projects.

TEACHING EXPERIENCE

Lancaster University

GRADUATE TEACHING ASSISTANT (GTA)

Lancaster, UK

October 2023 – December 2025

- Assisted in teaching undergraduate courses: Discrete Mathematics (MATH111), Numbers and Relations (MATH112), Real Analysis (MATH210), and Complex Analysis (MATH220).
- Supported student learning through grading, tutoring, and leading interactive discussion sections.
- Contributed to the development of course materials and fostered student engagement in problem-solving sessions.

Languages

Languages English (Full Professional Proficiency), Spanish (Native), German (basic)

Honors & Awards

- 2025 **Spring Into Quant Finance Programme**, G-Research
- 2024&2026 **Pure Mathematics Prize**, Lancaster University
- 2023 **1912 Senior Scholarship**, Fitzwilliam College, University of Cambridge
- 2023 **Fitzwilliam College Prize**, Fitzwilliam College, University of Cambridge
- 2022 **Postgraduate Fellowship Abroad**, "la Caixa" Foundation
- 2022 **Bachelor's Degree Extraordinary Award**, Universidad Complutense de Madrid (Ranked 1st in class)
- 2022 **Department Research Grant**, Dept. of Mathematical Analysis, UCM
- 2022 **Blas Cabrera Summer Scholarship**, Spanish Ministry of Science and Innovation
- 2022 **Arturo Duperier Award**, Pedro Bernardo City Council (Physics excellence)
- 2018–2021 **Excellence Grants**, Madrid City Council (Awarded 4 consecutive years)
- 2020 **Becas Atrévete 2020**, McKinsey & Company
- 2019 **California - UCM Scholarship**, Universidad Complutense de Madrid (Top 7 students selected)
- 2019 **Santander Progress Scholarship**, Santander Bank

Publications

JOURNAL ARTICLES

- **A. Acuaviva**. *The class of Banach lattices is not primary*. *Forum of Mathematics, Sigma*, 14:e41 (2026).
- **A. Acuaviva** and A. B. Nasser. *Quantifying (non-) weak compactness of operators on AL - and $C(K)$ -spaces*. *Proceedings of the Edinburgh Mathematical Society*, 1–56 (2026).
- **A. Acuaviva**. *Factorizations and minimality of the Calkin Algebra norm for $C(K)$ -spaces*. *Journal of the London Mathematical Society*, 112:e70321 (2025).
- **A. Acuaviva**. *Complemented subspaces of a $C(K)$ -space constructed by Candido*. *Journal of Mathematical Analysis and Applications*, 556:130115 (2026).
- **A. Acuaviva**. *Operators on injective tensor products of separable Banach spaces and spaces with few operators*. *Journal of Functional Analysis*, 290:111203 (2026).
- **A. Acuaviva**, A. Chan, S. Eldridge, C. Godsil, M. How-Chun-Lun, C. Tamon, E. Wright, and X. Zhang. *State Transfer in Chiral Quantum Walks*. *Philosophical Transactions A: Mathematical, Physical and Engineering Sciences*, 383 (2025).
- **A. Acuaviva** and D. Seco. *Equidistribution of zeros of some polynomials related to cyclic functions*. *Analysis and Mathematical Physics*, 12:53 (2022).

CONFERENCE PROCEEDINGS

- **A. Acuaviva**, J. Bautista, W. Yao, J. Jimenez, and H. Garcia de Marina. *Resilient Source Seeking with Robot Swarms*. In *Proceedings of the IEEE 63rd Conference on Decision and Control (CDC)*, Milan, Italy, pp. 57–63 (2024).

PREPRINTS & SUBMITTED WORK

- **A. Acuaviva** and P. Acuaviva. *Mathematical Discovery in the Wild: AI-Guided Proofs in Banach Space Theory*. arXiv:2607.17388 (2026).
- **A. Acuaviva** and P. Acuaviva. *A Unital Banach Algebra Which Is Not a Calkin Algebra*. arXiv:2607.15075 (2026).
- **A. Acuaviva**. *A Separable Banach Space with a Schauder Basis Which Is Not a Lipschitz Retract of Its Bidual*. arXiv:2607.12935 (2026).
- **A. Acuaviva**, B. Horváth, and T. Kania. *Pure infiniteness and primary factorisation*. arXiv:2607.01467 (2026).
- **A. Acuaviva**. *Preservation of primariness under ℓ_1 -, c_0 -, and ℓ_∞ -sums of Banach spaces*. arXiv:2606.26417 (2026).
- **A. Acuaviva**. *Primariness of the spaces $\ell_p(C(K))$ for $1 \leq p \leq \infty$* . arXiv:2605.29854 (2026).
- **A. Acuaviva** and T. Kania. *Primariness and the Primary Factorisation Property*. arXiv:2605.21711 (2026).
- **A. Acuaviva**. *$C(K)$ -spaces with few operators relative to posets*. arXiv:2511.22339 (2025).